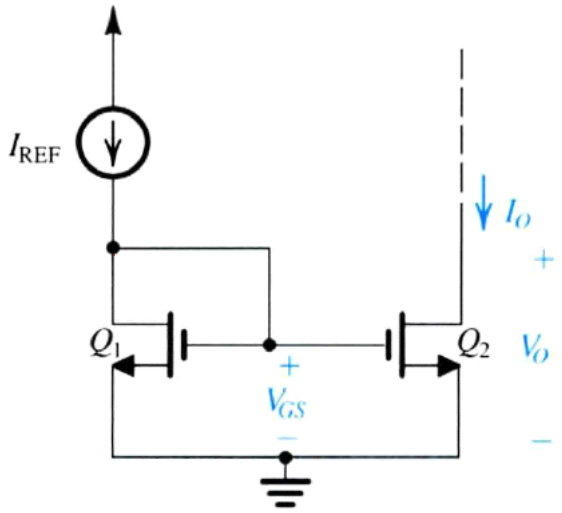
Reading

- Chapter 8.7 Current Mirror Circuits with Improved Performance
- Chapter 9.1 The MOS Differential Pair

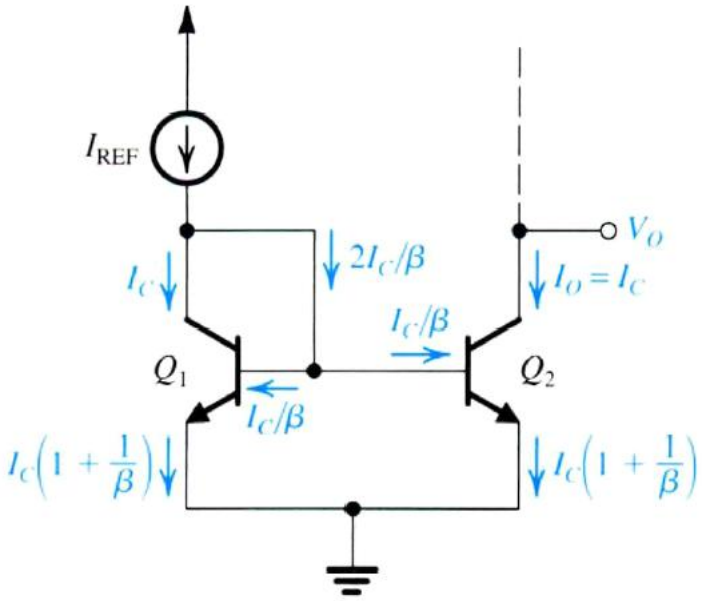
Recap (Current Sources)



$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

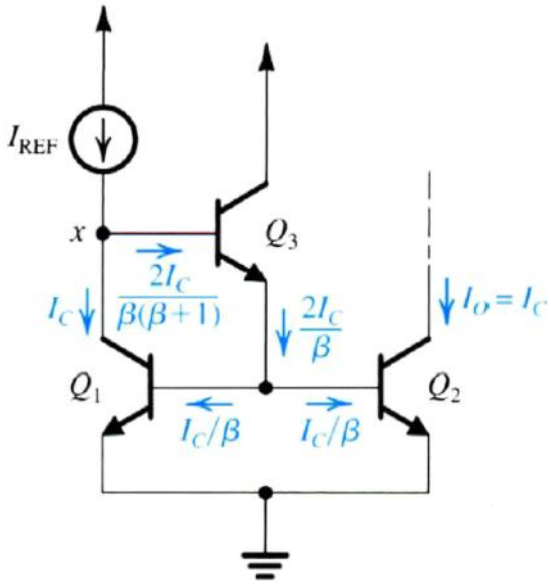
$$R_o = r_{o2}$$



$$\frac{I_O}{I_{REF}} = \frac{I_C}{I_C\left(1+\frac{2}{\beta}\right)} = \frac{1}{1+\frac{2}{\beta}}$$

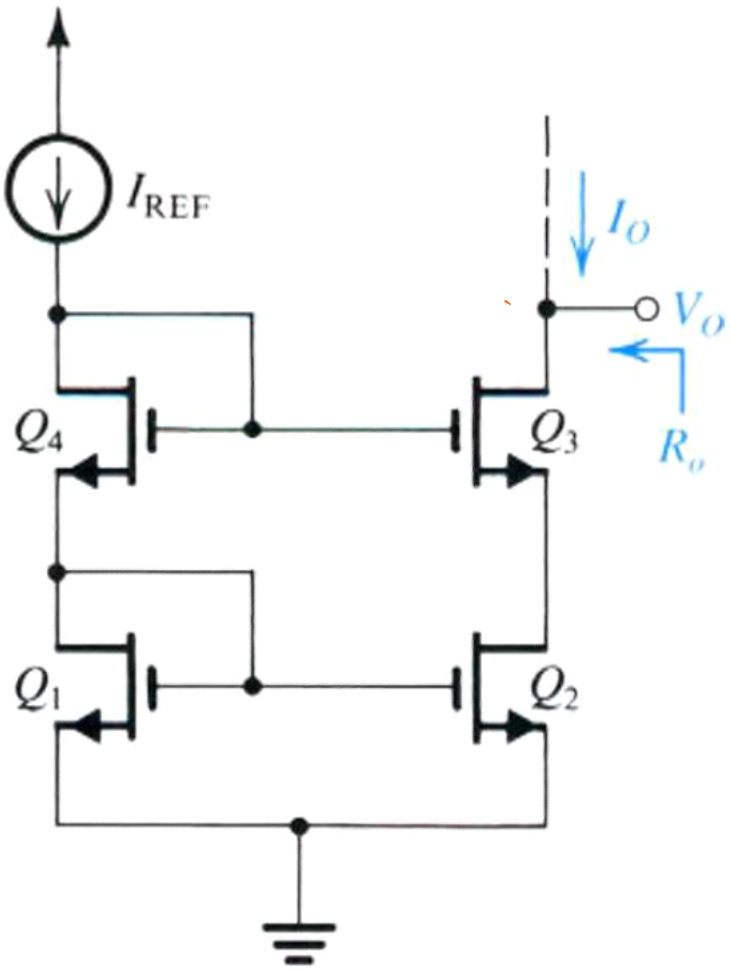
$$\frac{I_O}{I_{REF}} = \frac{m}{1+\frac{m+1}{\beta}}$$

$$I_O = I_{REF}\frac{m}{1+\frac{m+1}{\beta}}\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$



$$\frac{I_O}{I_{REF}} = \frac{1}{1+2/(\beta^2+\beta)} \simeq \frac{1}{1+2/\beta^2}$$

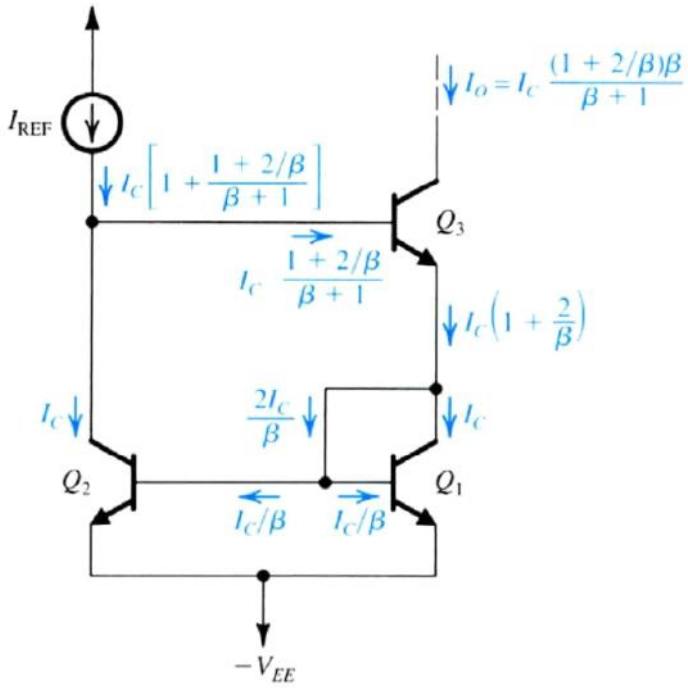
The Cascode Current Mirror



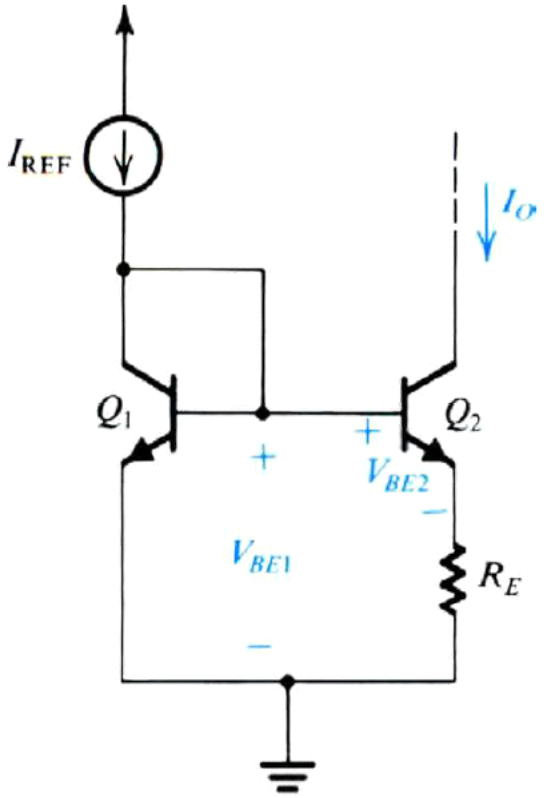
The Wilson MOS Mirror

The Wilson MOS Mirror

The Wilson BJT Current Mirror



The Widlar Current Source



Recap (OpAmp)

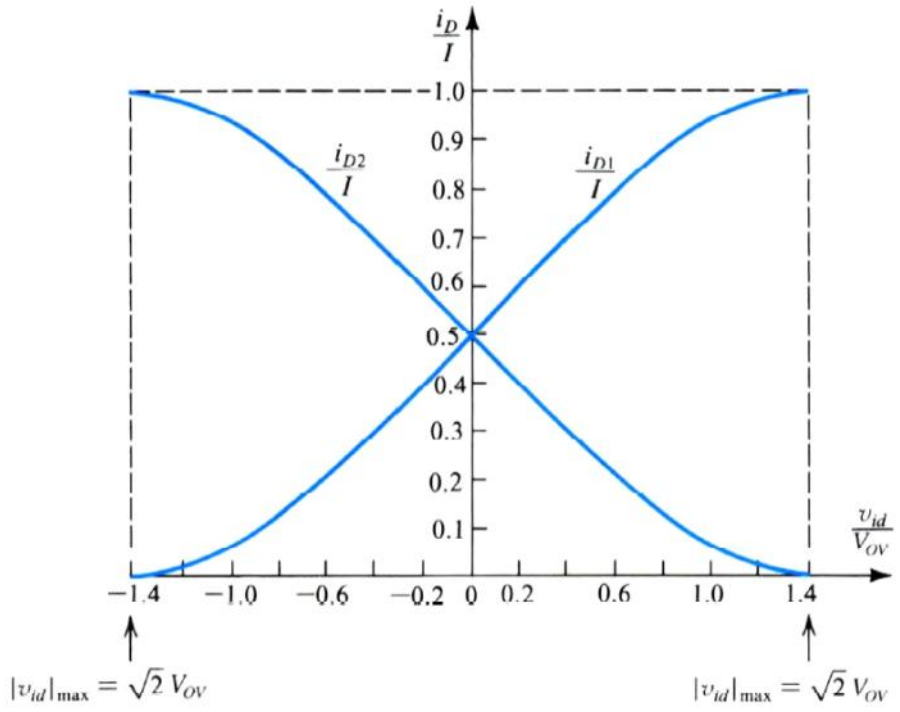
MOS Differential Pair with Common Input

MOS Differential Pair with Differential Input

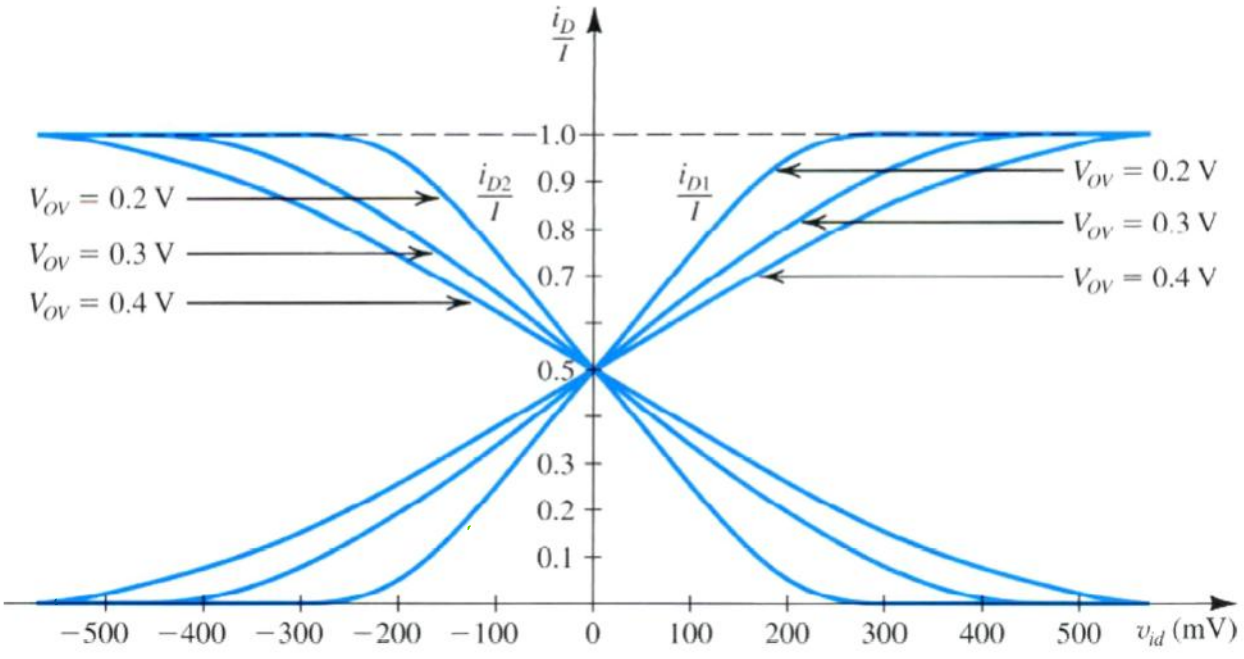
MOS Differential Pair with Differential Input

Large Signal Operation

Large Signal Operation



Large Signal Operation



Template